

DEVA ANAND

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Education

MS, Computer Science , University of Massachusetts Amherst	2025–2027
<i>Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering</i>	
BE, Computer Science and Engineering , Vels Institute of Science, Technology and Advanced Studies	2019–2023

Professional Experience

TherAlign Health , Founding Software Engineer Intern <i>Remote</i>	Jul 2026 – Present
- Founding engineer on a provider-facing SMART on FHIR clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization. - Extending a Firebase Cloud Functions backend that builds alternatives from RxNorm normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with Gemini constrained to synthesis while the backend deterministically controls eligibility and coverage.	
Culture & Morality Lab , UMass Amherst, Lab Assistant <i>Amherst, MA</i>	Jul 2026 – Present
- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, FastAPI, sentence-transformers, React), productionizing a published NLP method into a well-architected self-serve web tool for researchers. - Built a YAML model registry, a versioned construct library (99 constructs from 38 questionnaires), and a structured data-quality warning system; grew the backend suite to 40 tests (hermetic CI).	
Wysa , Backend Engineer <i>Bengaluru, India</i>	Jul 2023 – Jul 2025
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for 8+ UK clinical clients, processing 10,000+ monthly triage submissions via Mayden iaptus REST integrations; contributed to \$340K+ in revenue. - Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by 100+ hours/month and lifting labeling throughput by 60%. - Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.	

Projects

NEON Quantized GEMM – INT8/INT4 SIMD Kernels (GitHub)	C++17, ARM NEON, INT8/INT4 quantization, roofline
Built INT8/INT4 GEMM kernels in C++ for the ARM NEON <code>sdot</code> instruction, taking a naive fp32 baseline (3 GFLOP/s) to a 4×4 register-blocked int8 micro-kernel at 394 GFLOP/s (132× over naive, ~2.5× over the compiler's auto-vectorized code) by restructuring memory layout and hiding instruction latency with multiple accumulators; verified bit-exact against a scalar reference and benchmarked with p50/p99/p999 tail-latency methodology.	
Spark ETL Performance Optimization (GitHub)	PySpark, Spark UI, SQL, NYC TLC Dataset
Cut end-to-end Spark ETL runtime ~25% (17–19 min → 14.3 min) on a ~1.4B-row NYC TLC dataset via broadcast joins (shuffle minutes → ~0.05s), AQE and skew-join handling, and partition tuning; reduced the analytics tail-latency ratio (P99/P50) 2.14x → 1.74x.	
AgenticSearch – Provenance-First Entity Discovery (GitHub Demo)	Python, FastAPI, OpenAI / Groq, SQLite
Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and cell-level provenance; hardened with 150+ tests.	

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, Java, TypeScript, JavaScript, SQL, Bash, MongoDB, PostgreSQL, SQLite
Systems & Perf: SIMD intrinsics (AVX/NEON), roofline analysis, cache-aware tiling, Linux perf, Cachegrind, PySpark, Spark UI
Frameworks / Tools: FastAPI, Node.js, Express, React, Docker, Git, GitHub Actions, REST APIs, pytest
ML & AI: PyTorch, quantization (INT8/INT4), LLM agents, RAG, OpenAI / Groq APIs, sentence-transformers
Cloud / Platforms: AWS (KMS, S3), GCP, Azure, Linux, MongoDB Atlas

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.